THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS CHAPTER TEST SET A

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Virahāṅka sequence 1, 2, 3, 5, 8, … the term after 13 is: 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 26

2. The sum of 7 and 8 is: 1

(a) even **(b)** odd **(c)** neither **(d)** both

3. The sum of 45 and 12 is: 1

(a) odd **(b)** even **(c)** neither **(d)** both

4. The product 13 × 11 is: 1

(a) odd **(b)** even **(c)** neither **(d)** both

5. Which number is NOT a palindrome? 1

(a) 909 **(b)** 1331 **(c)** 1234 **(d)** 4554

6. A number is divisible by 5 if its last digit is: 1

(a) 5 only **(b)** 0 or 5 **(c)** any odd digit **(d)** 0 only

7. The sum of the first n odd numbers is: 1

(a) n^2 **(b)** n + 1 **(c)** 2n **(d)** n(n + 1)

8. A letter puzzle: if 3A + 2 = 17 then A = 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

9. The Virahāṅka numbers were first studied in India while counting: 1

(a) planet orbits **(b)** coins

(c) triangles **(d)** rhythm patterns (long and short beats)

10. Reverse of 4,521 is: 1

(a) 1,452 **(b)** 2,541 **(c)** 1,254 **(d)** 5,412

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the rule for the sum of two odd numbers and give an example. 2

12. Is the product of two odd numbers odd or even? Explain. 2

13. Write the first 8 terms of the Virahāṅka sequence. 2

14. What is a palindrome? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the sum of three consecutive numbers is divisible by 3. 3

16. Show how a Virahāṅka number can count rhythms of 6 beats. 3

17. Take the number 3,524. Apply the Kaprekar step once. 3

OR

Solve the cryptarithm: A B + A B = B C where A, B, C digits. 3

18. Complete the 3 × 3 magic square with centre 5 whose first row is 2, _, 6. 3

OR

Is the number 7,284 divisible by 3 and 4? Show. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Write the first 10 terms of the Virahāṅka sequence. 5
(ii) Show each rule step.
(iii) Which term is 144?

OR

Case study: A poet makes lines using short (1-beat) and long (2-beat) syllables. 5
(i) How many lines of 4 beats?
(ii) List them.
(iii) How many for 5 beats?
(iv) Name the sequence.

20. (i) Prove that the sum of two odd numbers is even. 5
(ii) Prove that odd × odd is odd.
(iii) Use these to decide the parity of 7 × 9 + 11.

OR

(i) Apply the Kaprekar routine to 4,321. 5
(ii) Repeat until it stops.
(iii) What constant do you get?

END OF QUESTION PAPER — Total: 40 Marks

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